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TEACHING STATEMENT

My teaching practice is grounded in my dual identity as an electrical engineer and an active researcher in embedded intelligent systems, robotics, IoT, energy management, and autonomous systems. Across approximately eight years of university teaching, laboratory instruction, student mentoring, and graduation project supervision, I have developed a teaching philosophy that connects rigorous theoretical foundations with practical engineering implementation. I believe that engineering education should guide students from understanding concepts to designing, testing, validating, documenting, and communicating complete technical solutions.

My academic teaching experience includes real-time systems, embedded systems programming, Internet of Things, industrial computing, electronics, electrotechnics, automation, electrical engineering workshops, and biomedical measurement. These experiences were developed through teaching appointments at the Higher Institute of Technological Studies of Sidi Bouzid, the Higher Institute of Industrial Management of Sfax, and the Higher Institute of Biotechnology of Sfax. They have allowed me to teach students with diverse levels of preparation and to adapt my pedagogy to both classroom-based and laboratory-based learning environments.

TEACHING PHILOSOPHY

My main objective as a teacher is to prepare students to become independent, rigorous, creative, and ethically responsible engineers. I do not view teaching as the simple transfer of information. I view it as a structured learning process through which students learn how to reason, model, implement, debug, evaluate, and improve engineering systems. For this reason, I combine lectures, tutorials, laboratory work, simulations, mini-projects, and final-year projects. I encourage students to ask questions, test assumptions, analyze errors, and justify their technical choices using scientific reasoning.

In technical courses, I begin with the fundamental principles and progressively connect them to real applications. For example, in embedded systems and IoT, students first study microcontroller architecture, sensors, actuators, communication protocols, and data acquisition. They then implement prototypes using platforms such as Arduino, Raspberry Pi, STM32, ESP32, and similar embedded boards. In automation and electrotechnics, I connect theoretical models to laboratory measurements, wiring practice, control logic, safety requirements, and troubleshooting procedures. This approach helps students understand that engineering knowledge becomes meaningful when it is verified through implementation and experimental validation.

I also emphasize the importance of learning by doing. Students are encouraged to make technical decisions, compare different implementation choices, evaluate limitations, and interpret unexpected results. This process builds confidence and transforms practical sessions from procedural exercises into opportunities for engineering reasoning.

TEACHING AREAS AND LABORATORY EXPERIENCE

Teaching area	Courses and laboratory experience
Embedded systems and IoT	Embedded systems programming, sensor interfacing, microcontroller-based control, real-time data acquisition, communication protocols, and IoT-based monitoring.
Real-time and industrial computing	Real-time systems, industrial computing workshops, algorithm implementation, timing constraints, and software-hardware interaction.
Electronics and electrotechnics	Electrical engineering workshops, electronic circuits, electrotechnical systems, measurement instruments, wiring, and practical validation.
Automation and control	Basics of automation, Grafcet logic, control of industrial systems, system diagnosis, and laboratory troubleshooting.
Biomedical measurement	Biomedical measurement using the KL-72001 platform, signal acquisition, instrumentation, and practical interpretation of measured data.
Renewable energy and power systems	Photovoltaic systems, energy conversion, battery-related applications, measurement, and laboratory-based analysis of electrical systems.

My laboratory teaching is based on progressive autonomy. At the beginning of a practical session, I provide the scientific background, objectives, safety instructions, expected outcomes, and evaluation criteria. During the session, students are guided to connect components, configure software tools, acquire measurements, interpret results, and identify possible sources of error. By the end of the session, they are expected to present their work in a technical report that includes the objective, methodology, experimental setup, results, discussion, and conclusion.

PROJECT-BASED LEARNING AND STUDENT SUPERVISION

Project-based learning is a central element of my teaching practice. I have supervised and co-supervised graduation projects in IoT, artificial intelligence, robotics, photovoltaic systems, electronics, automation, and embedded systems. These projects include intelligent airport intrusion monitoring using IoT and AI, the design and realization of a connected mailbox, surveillance and security drones, Arduino-based acquisition systems for photovoltaic panels, electronic cards for monitoring physical quantities in photovoltaic installations, and charge regulator design for autonomous photovoltaic systems.

During supervision, I guide students through the full engineering development cycle. This includes defining the problem, identifying system requirements, reviewing the state of the art, selecting components, designing hardware or software architectures, implementing prototypes, testing performance, interpreting results, and preparing written and oral presentations. I place strong emphasis on clear methodology, reproducibility, technical documentation, and professional communication. My role is to help students become autonomous while ensuring that their work remains scientifically coherent and technically feasible.

- I help students transform broad project ideas into precise engineering objectives, functional specifications, and measurable deliverables.
- I encourage the use of block diagrams, flowcharts, experimental protocols, validation tables, and clear result interpretation.
- I support students in writing structured reports and preparing professional presentations for academic juries.

- I promote teamwork, responsibility, ethical engineering practice, and respect for safety procedures in laboratory and project environments.

ASSESSMENT AND FEEDBACK

My assessment strategy combines continuous evaluation, laboratory performance, technical reporting, oral communication, and project outcomes. In practical courses, I evaluate not only the final result but also the student process: preparation, understanding of the problem, ability to use instruments, implementation quality, interpretation of measurements, debugging capacity, and clarity of the report. This type of assessment helps students recognize that engineering quality depends on both the final prototype and the reasoning behind its design.

Feedback is an essential part of my teaching. I provide comments that are specific, constructive, and focused on improvement. When students make mistakes, I use them as learning opportunities by asking them to identify the cause, propose corrections, and compare expected and observed results. This method strengthens their technical confidence and develops a problem-solving mindset. I also encourage students to improve their scientific writing, graphical presentation of results, and oral explanation of technical decisions.

In laboratory and project modules, I value progress as much as outcomes. I therefore encourage students to document intermediate results, record failed attempts, justify design changes, and explain the limitations of their work. This practice prepares them for research, engineering development, and professional reporting.

INTEGRATION OF RESEARCH INTO TEACHING

My teaching is strongly enriched by my research activities in UAV navigation, robotics, embedded artificial intelligence, energy management, battery health monitoring, and IoT-based intelligent systems. I use examples from my research to show students how theoretical concepts are applied in modern engineering problems. For instance, sensor fusion concepts can be explained through UAV localization in GNSS-denied environments; embedded programming can be connected to real-time robotics applications; and energy conversion concepts can be linked to hybrid electric vehicles, batteries, supercapacitors, and renewable energy systems.

This research-informed teaching approach helps students see the connection between academic courses and real technological challenges. It also motivates them to explore interdisciplinary topics that combine electrical engineering, computer science, artificial intelligence, robotics, and energy systems. My long-term objective is to develop teaching materials and student projects that reflect current advances in embedded AI, autonomous systems, smart mobility, and intelligent sensing.

MENTORING, INCLUSION, AND LEARNING ENVIRONMENT

I aim to create a respectful, motivating, and inclusive classroom environment where students feel comfortable asking questions and developing their ideas. I recognize that students do not all learn in the same way; therefore, I use multiple forms of explanation, including mathematical derivations, block diagrams, simulations, hardware demonstrations, code examples, and practical case studies. I also encourage peer learning, discussion, and collaborative problem solving, especially in laboratory and project-based activities.

Mentoring is one of the most meaningful aspects of my teaching role. I support students not only in technical development but also in building confidence, discipline, scientific curiosity, and professional behavior. Many students need guidance in organizing their work, managing time, writing reports, and preparing for jury evaluation. I consider these skills essential for their transition from students to engineers and researchers.

Because my teaching experience spans different institutions and student profiles, I have learned to adapt my explanations, examples, and levels of guidance according to the needs of each group. My goal is to maintain high academic expectations while providing the structure and support needed for students to succeed.

FUTURE TEACHING GOALS

In the future, I aim to further develop modern and research-oriented teaching modules in embedded systems, robotics, IoT, artificial intelligence, renewable energy, and autonomous systems. I am particularly interested in creating laboratory activities that integrate ROS 2, UAV simulation, embedded AI, sensor fusion, smart monitoring, and energy management. I also plan to continue improving my teaching materials by incorporating clear diagrams, structured laboratory guides, real datasets, and open-source tools that allow students to reproduce and extend their experiments.

My overall goal is to prepare students who can combine theory, practice, creativity, and ethical responsibility. I want my students to leave my courses with the ability to analyze complex systems, implement reliable solutions, communicate their results clearly, and continue learning independently. Through teaching, supervision, and mentoring, I seek to contribute to the development of engineers who are technically strong, research-aware, and ready to address real-world challenges in intelligent systems, energy, robotics, and digital transformation.